

CSCI 49900: Abstract

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Large language models are increasingly ubiquitous in many aspects of industry, academia, and research, and are easily accessible to non-specialized consumers for personal use. Yet, the high level of complexity of such models means that the mechanics of model inference are difficult for users to understand, observe directly, or experiment with. Currently, exploring the role of specific internal components, as well as the impact of optimization methods on model responses and performance, often requires specialized knowledge, tools, or programming experience.

The Language Model Lab project is an interactive platform to demonstrate the mechanics of large language model inference optimization and to allow for the exploration of model behavior through internal interventions, without the need for special tools or programming.

One part of our project is building a custom, optimized engine that allows a small, pre-trained AI model to potentially run faster and use less memory by representing its internal parameters at lower numerical precision, a process known as quantization. This optimization process will also involve writing a custom CUDA kernel, a GPU-executed function designed to accelerate computations involved in language model inference, with the aim of improving baseline implementations.

Another aspect of our project will involve creating a backend that takes parameters from the user, allowing them to perform interventions on the model, with the aim of comparing responses and exploring how internal components affect model behavior. Some interventions will center on decoding parameters affecting token selection, like temperature, top-k, top-p, and repetition penalty. Others will be model-internal interventions, which affect inner computations before next-token probabilities are produced. Examples of these are ablating multi-layer perceptron units in specific layers, ablating attention heads, or modifying weights.

To integrate these aspects of the project, we will design a user-facing web application. It will showcase how our quantization and kernel optimizations affect inference performance through the display of various metrics, and will also allow users to interact with the model. Users will be able to perform interventions of different kinds, and observe how the model's behavior changes compared to a baseline, depending on the intervention performed. The application will have informative graphics, explanations, and easy-to-use controls, helping users develop a deeper understanding of language model optimization, architecture, and behavior.